

worth the effort. In the initial days when we had just two sites, we hosted them on Rackspace because it had a 256 MB offering, which we found very attractive. However, when most of our visitors (at least 99 per cent of them were from India) complained of slow response times, which was due to the network latency, we tried the CDN (Cloudflare) route as a quick fix solution and the problem got compounded. The CDN wouldn't refresh content and visitors were being served stale pages. I suggest that, as a first step, you should move your hosting to a region where most of your traffic originates. For the others, consider implementing a reverse proxy like Squid or Varnish.

Use a reverse proxy

After Boost, adding a reverse proxy will give your site one of the biggest performance improvements. There are two candidates for this—Squid and Varnish. Though opinion on which one to use for Drupal sites is divided, I am in the process of configuring Varnish on one of the Rackspace servers in the UK. Varnish provides its own scripting language, VCL, and offers acceleration by a factor of 300 to 100 times.

Replace Apache with Nginx

Nginx is a high performance HTTP server. You could consider replacing Apache with Nginx. You should keep in mind that Boost has limited support for Nginx.

Specify an output handler for PHP

If you are constrained for bandwidth, specify the following directive in *php.ini*:

```
output_handler = ob_gzhandler
```

With this, output is compressed for browsers that accept compressed content.

Optimise PHP to Apache communication

To optimise PHP to Apache communication, set the directive in *php.ini* by uncommenting the following lines:

```
; output_buffering
; Default Value: Off
; Development Value: 4096
; Production Value: 4096
```

Use the PHP tidy extension

The PHP tidy extension is used to remove the white spaces in rendered HTML so that the payload sent to the browsers is smaller.

It is possible to increase the performance of a Drupal site without adding additional hardware by tuning the various components of the hosting stack, especially by configuring various caching mechanisms. 

References

- [1] CERT 7 Test Results <http://web.archive.org/web/20061115173834/http://sourcelabs.com/pdfs/SourceLabsApacheMySQLPHPTTestResults.pdf>
- [2] <http://tools.pingdom.com/fpl/> and <https://www.site24x7.com/>
- [3] <http://munin-monitoring.org/>
- [4] <http://foxfox.com/> (<https://addons.mozilla.org/en-US/firefox/addon/httpfox/>). Other tools that you could use are Httpie (<https://github.com/jkr/htp>), Fiddler (<http://www.fiddler2.com/fiddler2/>) or plain old Telnet.
- [5] <http://developer.yahoo.com/performance/rules.html>
- [6] Boost (<http://drupal.org/project/boost>)
- [7] Other Options: TUX - "Red Hat Content Accelerator" - <http://www.redhat.com/docs/manuals/tux/>, kHTTpd - <http://www.ferrus.demon.nl/>, httdpd - <http://www.acme.com/software/httdpd/>, lighthead () , lingersd (<http://www.iagora.com/about/software/lingersd/>)
- [8] <http://www.squid-cache.org/>
- [9] <http://varnish-cache.org/>
- [10] httdpd - <http://www.acme.com/software/httdpd/>, Caudium - <http://caudium.net/index.html>, Roxen - <http://www.roxen.com/products/webserver/>, Zeus (\$\$) - <http://www.zeus.co.uk>

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